



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

39K

Rows

Transaction records

18

Columns

Features per record

37

Missing Values

In Review Rating only

4

Categories

Clothing, Footwear, Accessories, Outerwear

Customer Demographics

Age, Gender, Location, Subscription Status

Purchase & Behavior

Item, Category, Amount, Season, Discount, Shipping, Review Rating,
Purchase Frequency

Data Preparation in Python

01

Load & Explore

Imported dataset via **pandas**; used `df.info()` and `.describe()` for initial structure and statistics review.

02

Clean & Standardize

Imputed missing Review Ratings using **median per category**; renamed columns to snake_case.

03

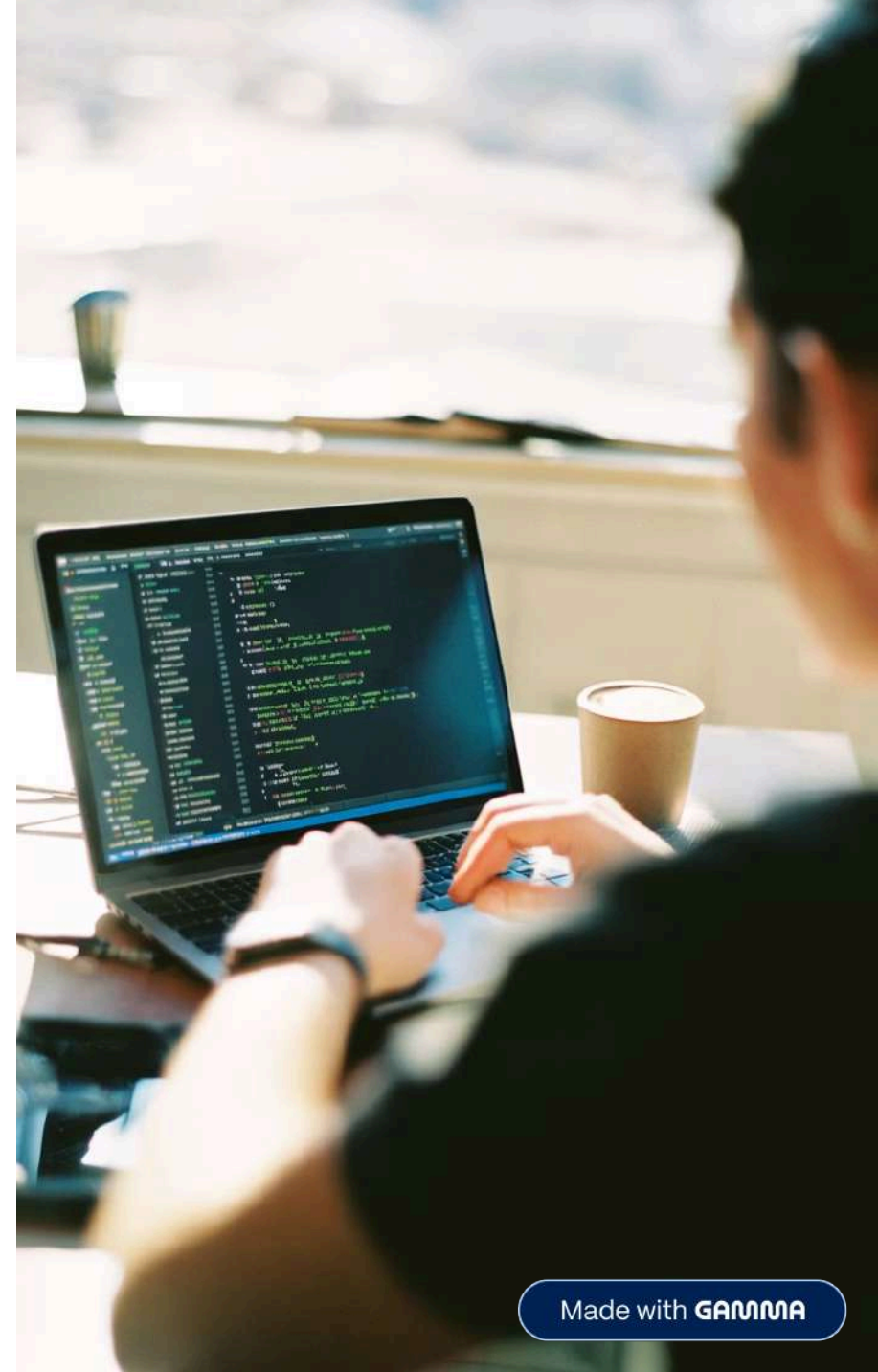
Feature Engineering

Created `age_group` and `purchase_frequency_days` columns; dropped redundant `promo_code_used`.

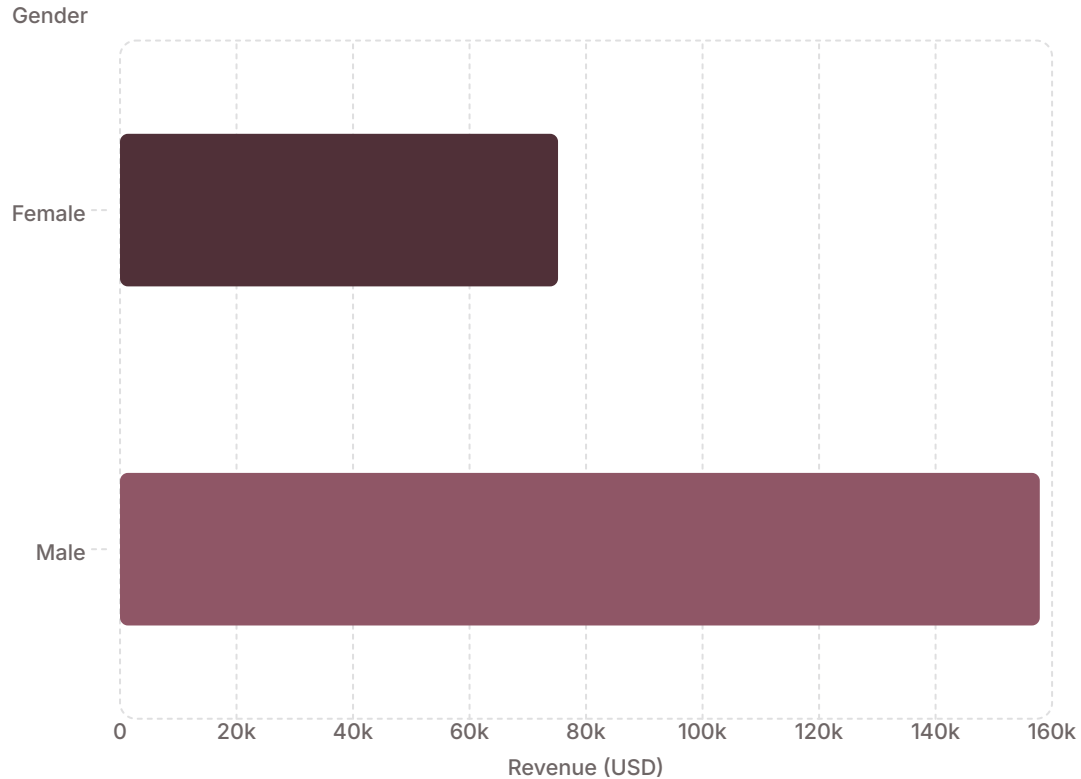
04

Database Integration

Loaded cleaned DataFrame into **PostgreSQL** for SQL-based analysis.



Revenue by Gender



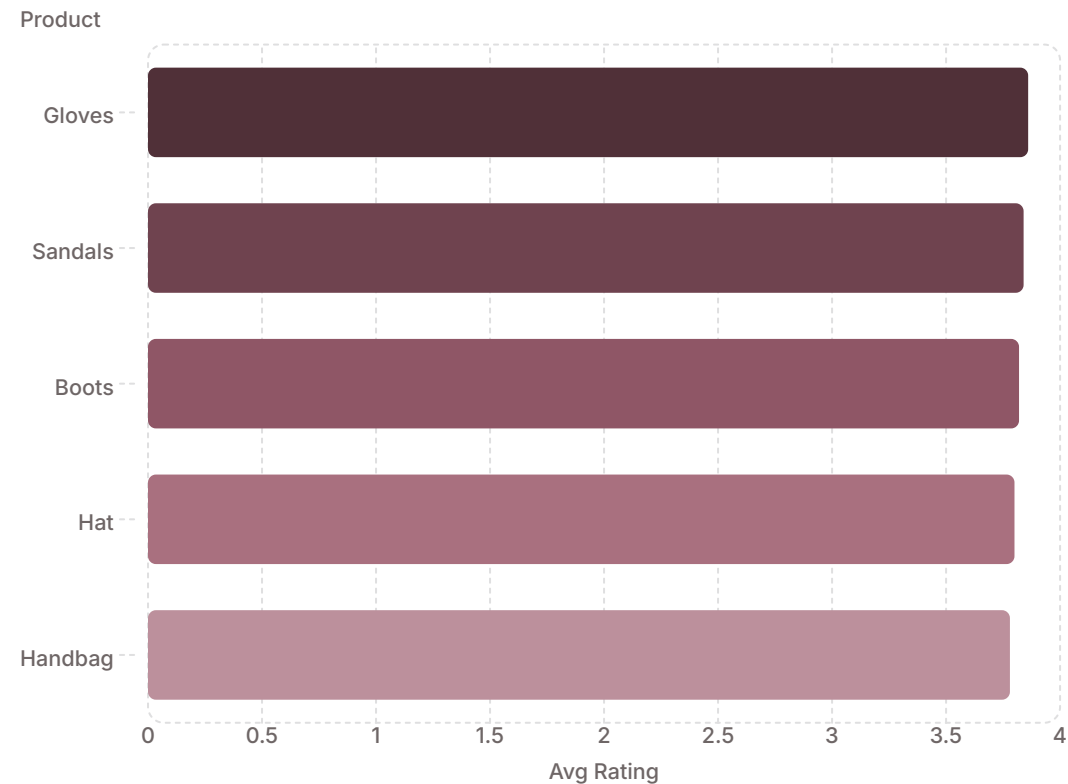
Key Insight

Male customers generated **more than double** the revenue of female customers — \$157,890 vs. \$75,191 — despite females representing a significant portion of the customer base.

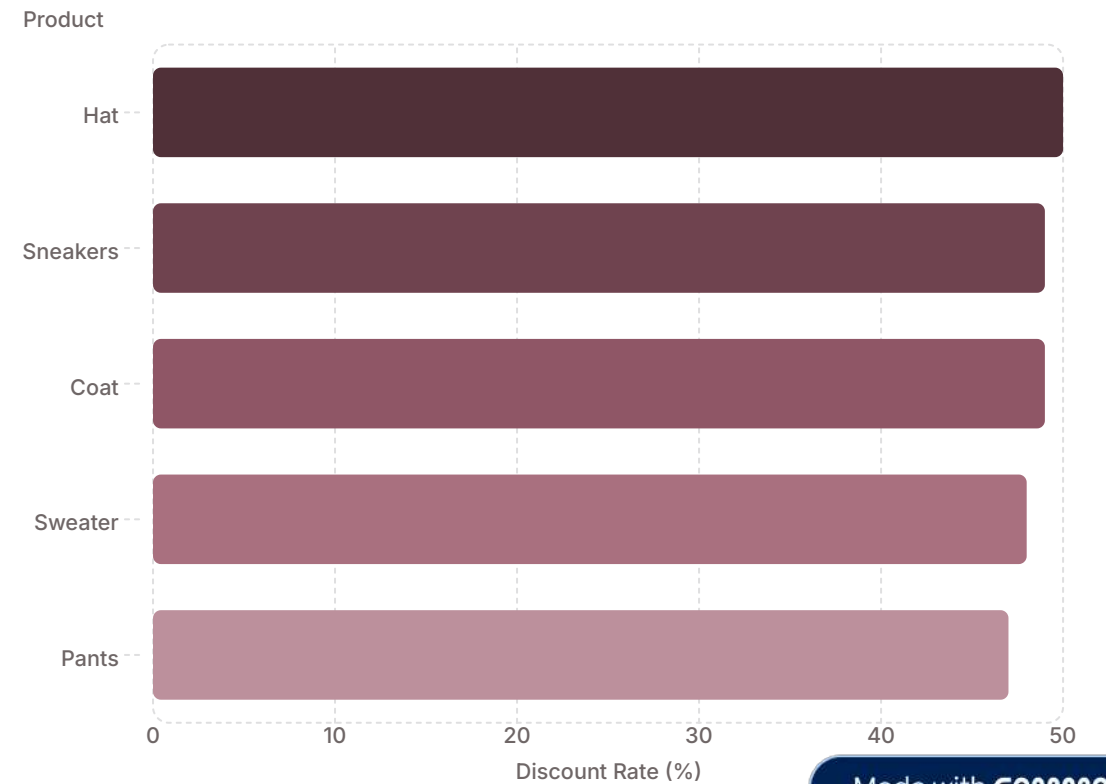
- ❏ Male customers account for 2,652 of 3,900 records, driving the revenue gap.

Top Products & Discount Patterns

Top 5 Products by Rating

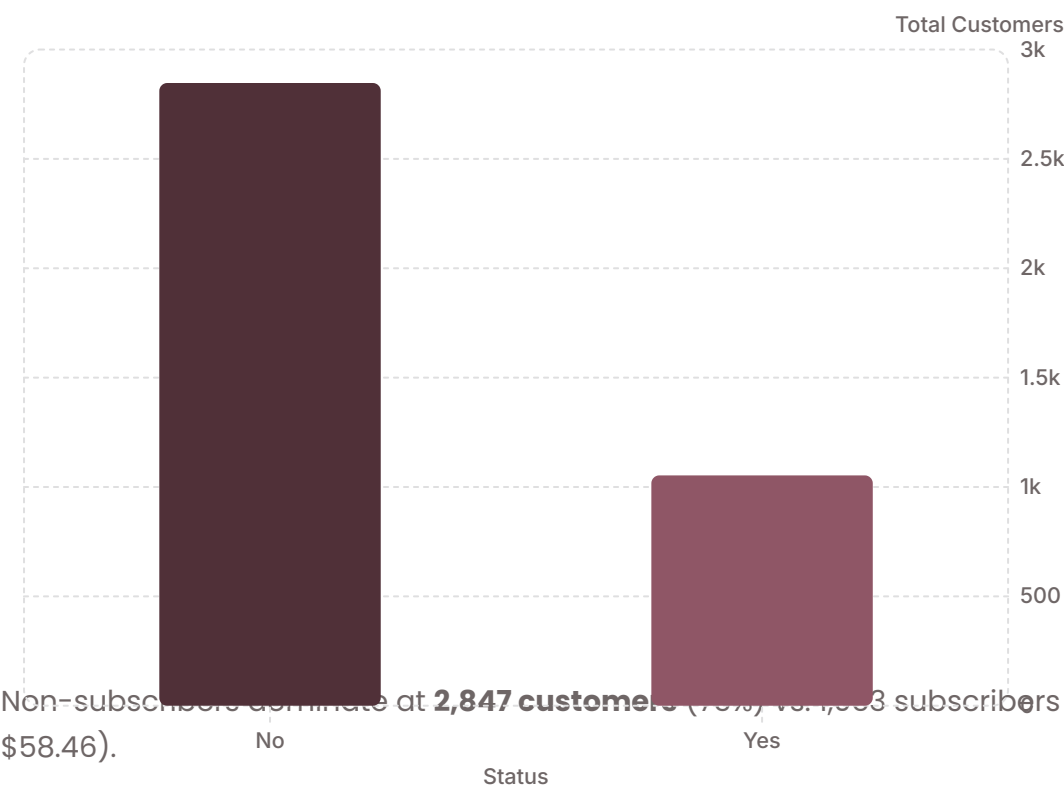


Most Discount-Dependent Products

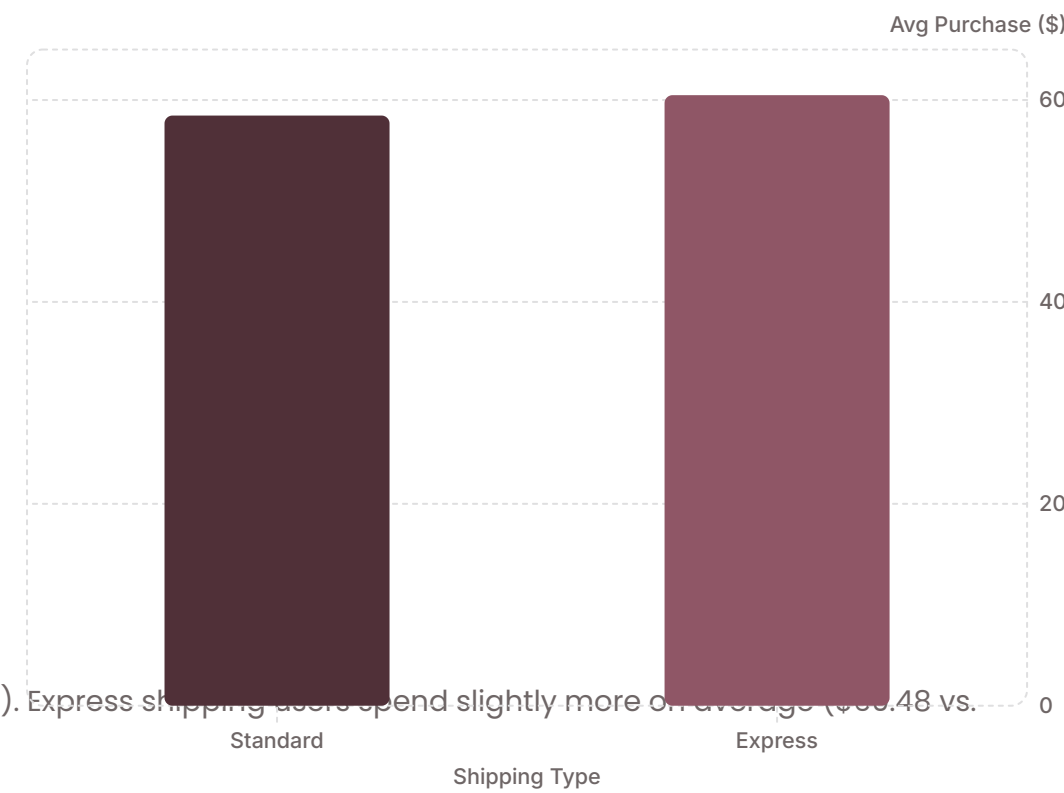


Subscription & Shipping Insights

Subscribers vs. Non-Subscribers



Average Spend by Shipping Type



Customer Segmentation & Age Groups

Loyal

3,116 customers — the dominant segment, built through repeat purchase history.

Returning

701 customers — engaged but not yet fully loyal; key conversion target.

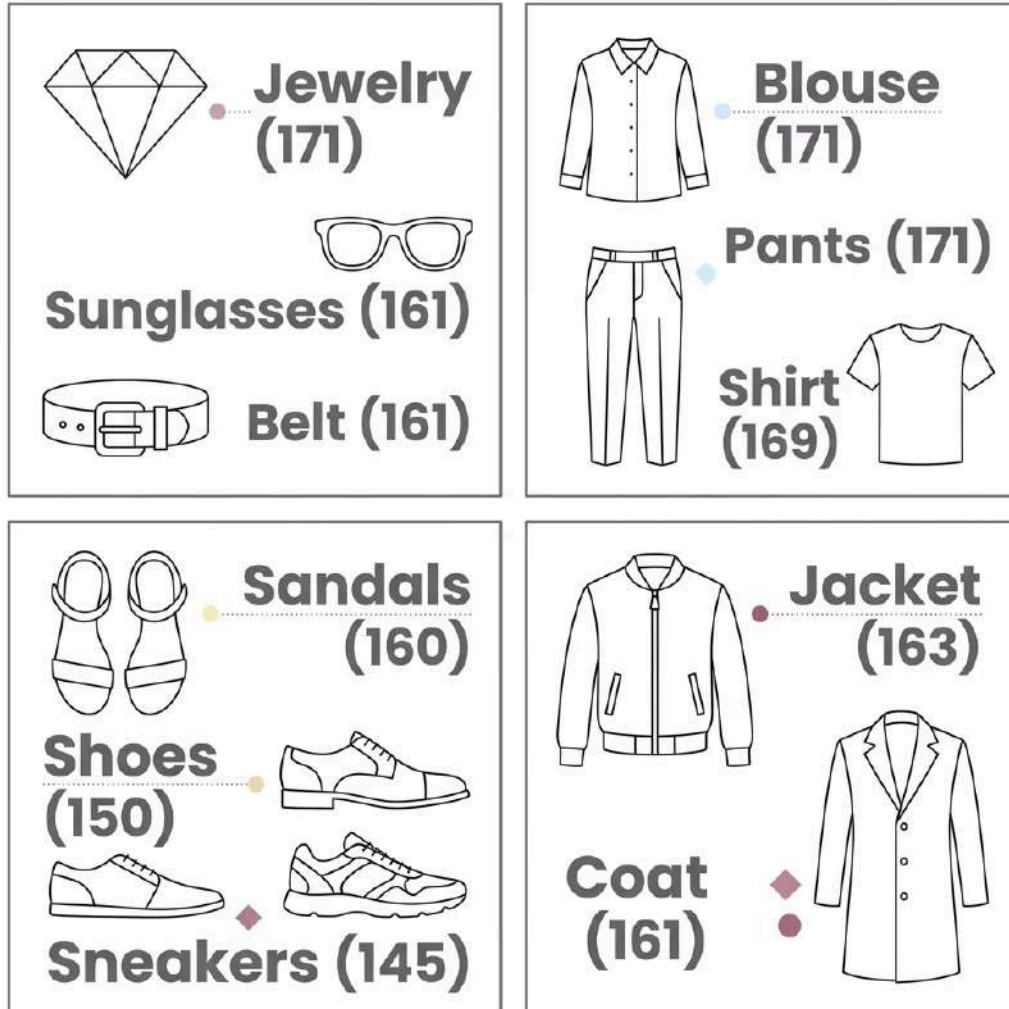
New

83 customers — smallest segment; opportunity for onboarding campaigns.

Revenue by age group: **Young Adults** lead at \$62,143, followed by Middle-aged (\$59,197), Adults (\$55,978), and Seniors (\$55,763).



Top Products by Category



Category Leaders

Clothing dominates overall with Blouse, Pants, and Shirt as top sellers. **Footwear** is led by Sandals (160 orders). **Outerwear** has only two top products listed: Jacket and Coat. **Accessories** rounds out with Jewelry leading at 171 orders.

Power BI Dashboard

KPIs

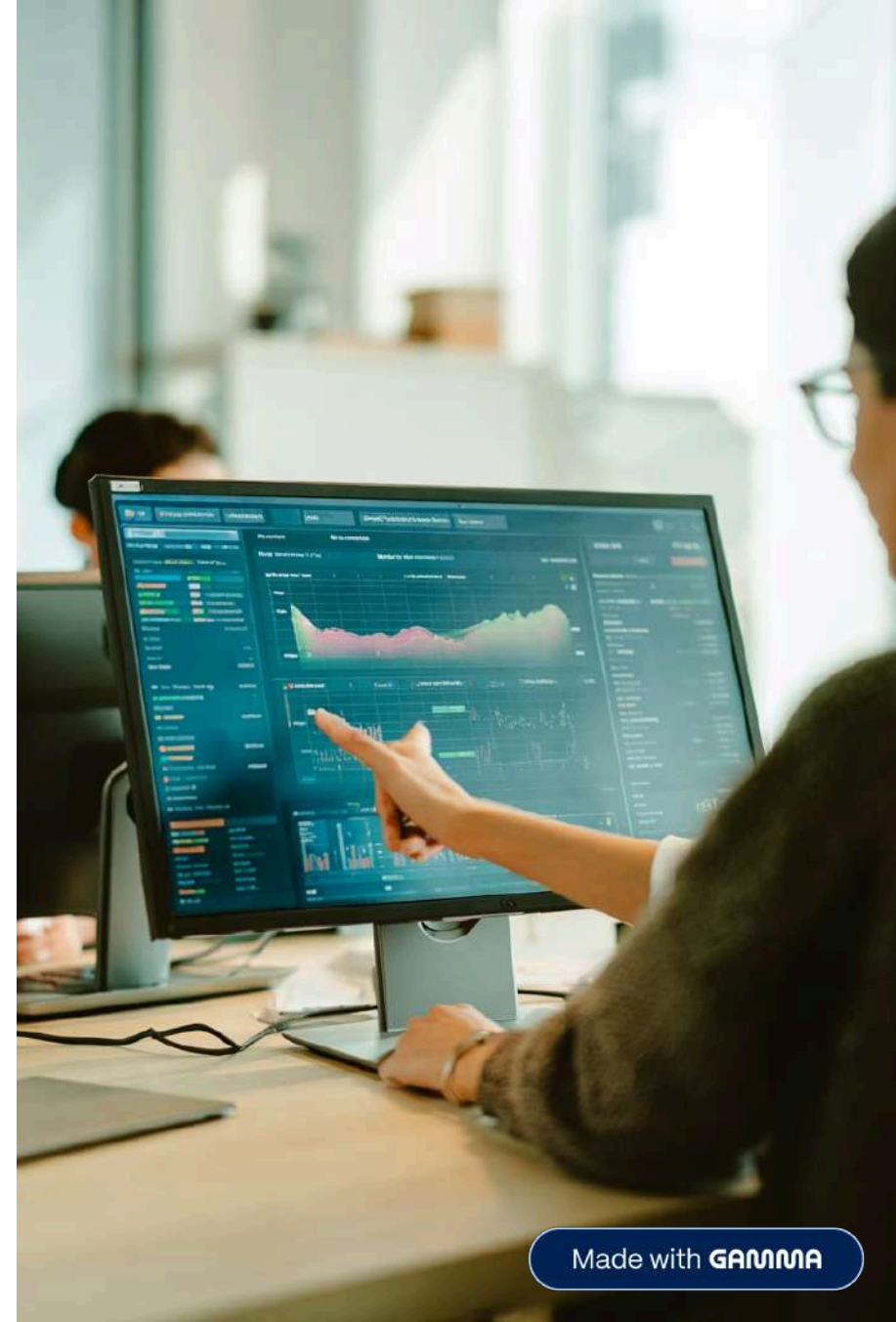
3.9K Total Customers · \$59.76 Avg Purchase · 3.75 Avg Review Rating

Filters

Subscription Status, Gender, Category, Shipping Type

Visualizations

Revenue & Sales by Category and Age Group; Subscription % donut chart



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.



Review Discount Policy

Balance sales boosts with margin control on high-discount products like Hats and Sneakers.



Loyalty Program

Reward repeat buyers to move Returning customers into the Loyal segment.



Targeted Marketing

Focus on high-revenue age groups (Young Adults) and Express-shipping users.